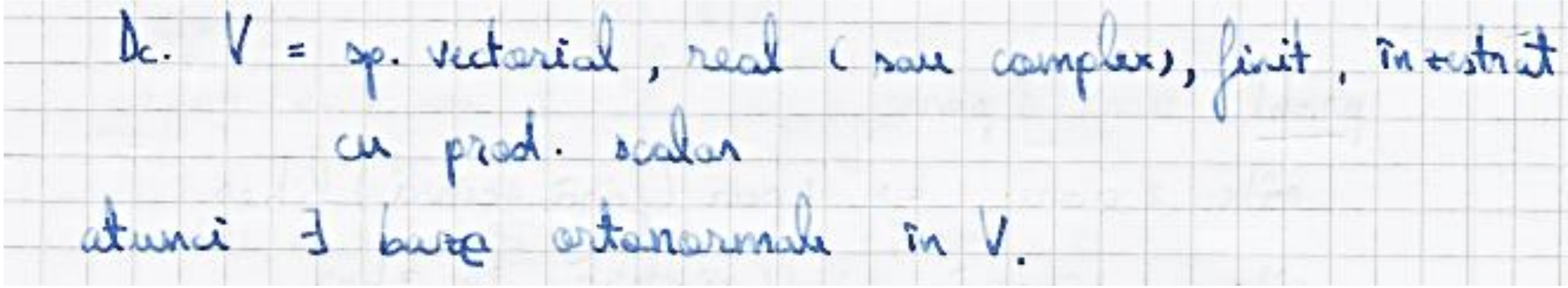
MATEMATICI 2

Curs 11

Procedeul Gram-Schmidt de ortonormare a unei baze

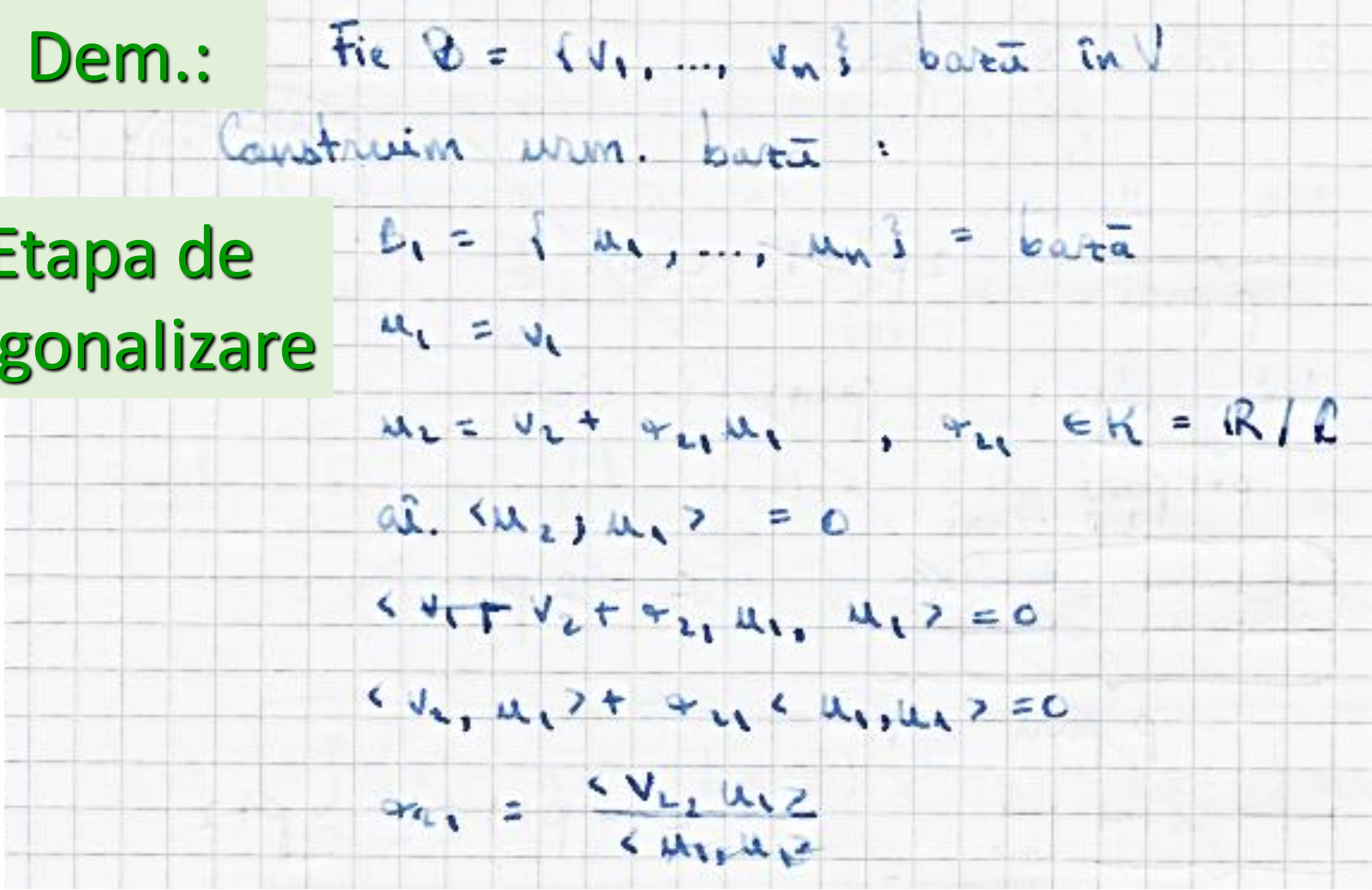


Dc. V = sp. vectorial, real (sau complex), finit, înzestrat
cu prod. scalar

atunci $\exists$ baza ortonormală în V.

Dem.:

I. Etapa de ortogonalizare



Fie $\mathcal{B} = \{v_1, \dots, v_n\}$ bază în V

Construim urm. bază:

$\mathcal{B}_1 = \{u_1, \dots, u_n\}$ = bază

$u_1 = v_1$

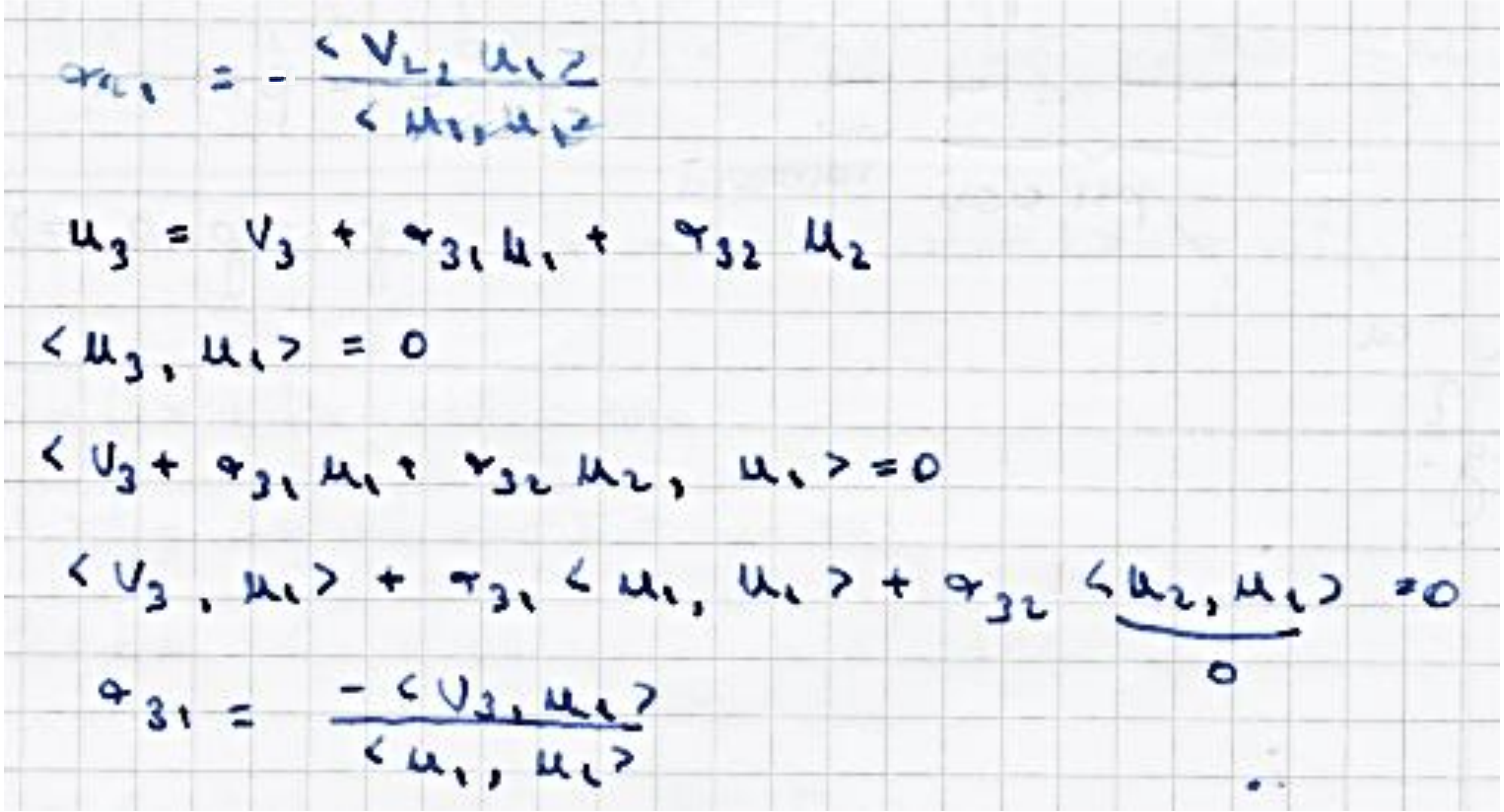
$u_2 = v_2 + \alpha_{21} u_1$, $\alpha_{21} \in K = \mathbb{R} / \mathbb{C}$

a.î. $\langle u_2, u_1 \rangle = 0$

$\langle \cancel{v_1} \; v_2 + \alpha_{21} u_1, u_1 \rangle = 0$

$\langle v_2, u_1 \rangle + \alpha_{21} \langle u_1, u_1 \rangle = 0$

$\alpha_{21} = \dfrac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle}$



$$\alpha_{21} = -\frac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle}$$

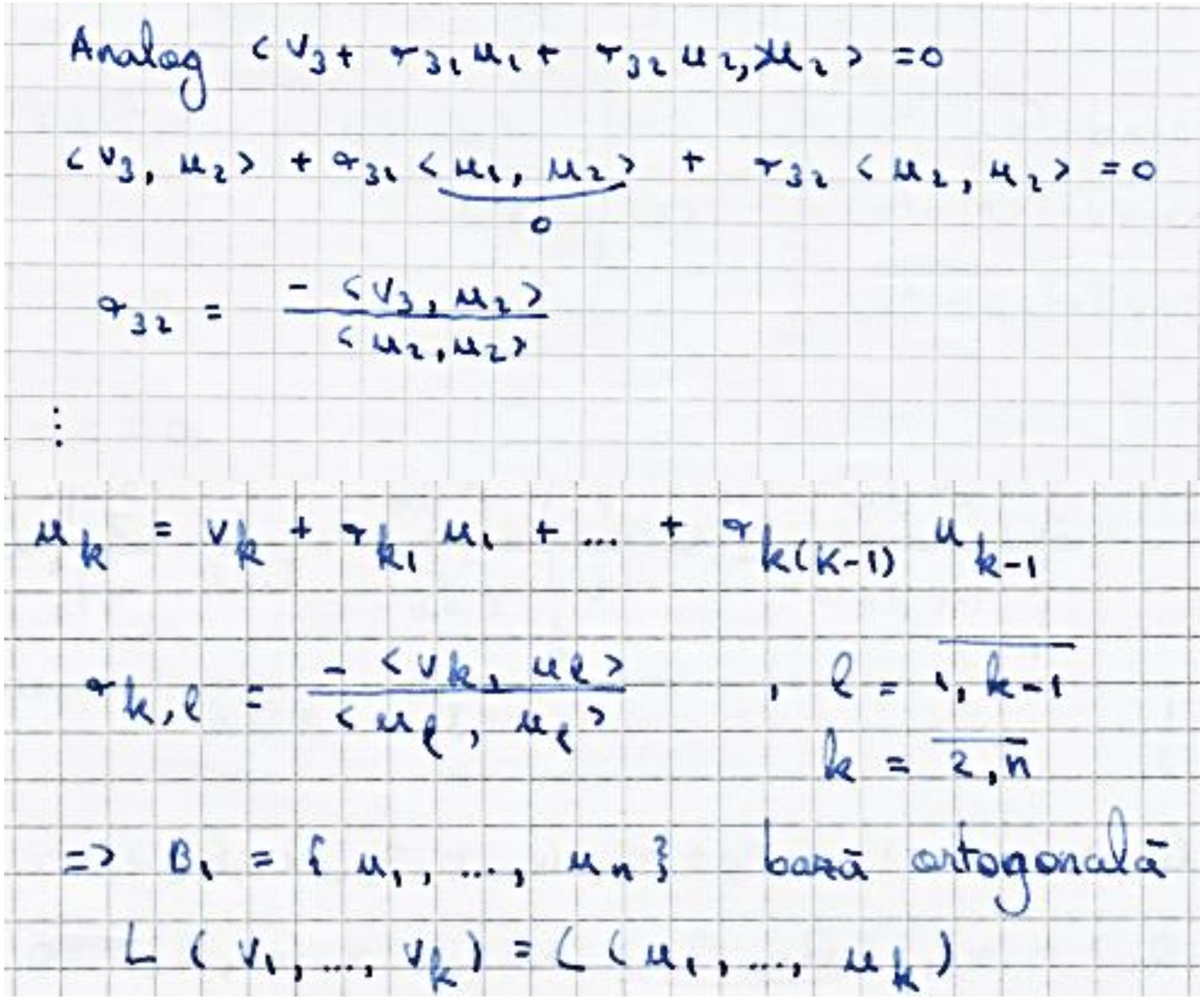
$$u_3 = v_3 + \alpha_{31} u_1 + \alpha_{32} u_2$$

$$\langle u_3, u_1 \rangle = 0$$

$$\langle v_3 + \alpha_{31} u_1 + \alpha_{32} u_2, \; u_1 \rangle = 0$$

$$\langle v_3, u_1 \rangle + \alpha_{31} \langle u_1, u_1 \rangle + \alpha_{32} \underbrace{\langle u_2, u_1 \rangle}_{0} = 0$$

$$\alpha_{31} = \frac{-\langle v_3, u_1 \rangle}{\langle u_1, u_1 \rangle}$$



Analog $\langle v_3 + r_{31}u_1 + r_{32}u_2, u_2\rangle = 0$

$$\langle v_3, u_2\rangle + r_{31}\underbrace{\langle u_1, u_2\rangle}_{0} + r_{32}\langle u_2, u_2\rangle = 0$$

$$r_{32} = \frac{-\langle v_3, u_2\rangle}{\langle u_2, u_2\rangle}$$

$\vdots$

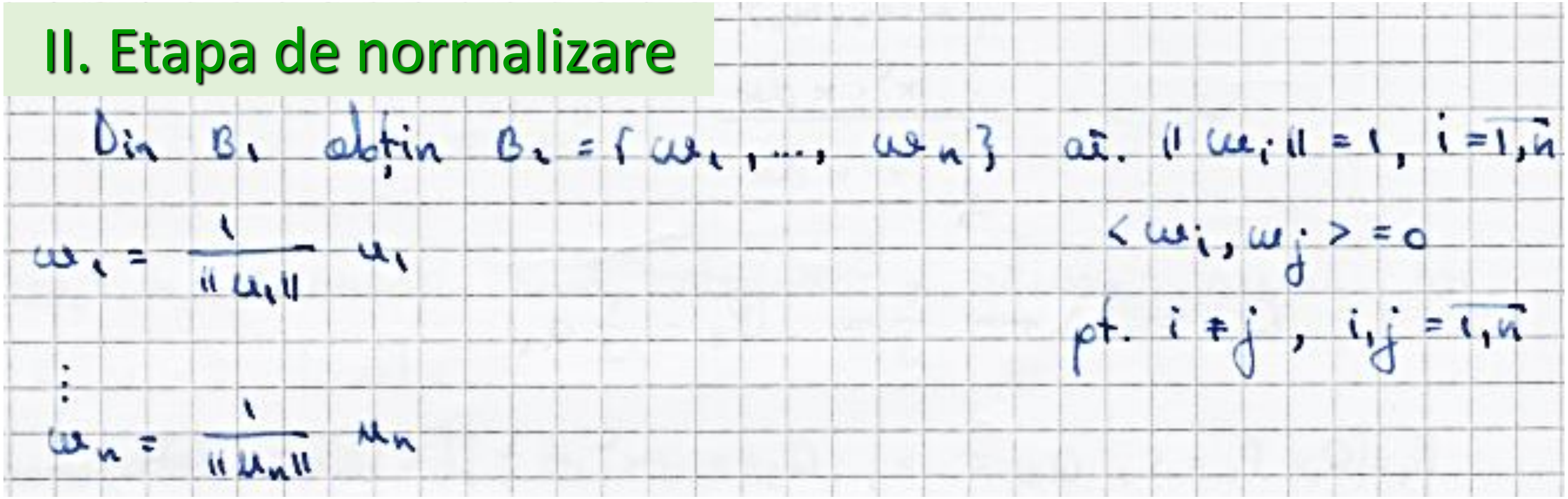
$$u_k = v_k + r_{k1}u_1 + \dots + r_{k(k-1)}u_{k-1}$$

$$r_{k,l} = \frac{-\langle v_k, u_l\rangle}{\langle u_l, u_l\rangle}, \quad l = \overline{1,k-1}$$
$$k = \overline{2,n}$$

$\Rightarrow B_1 = \{u_1, \dots, u_n\}$ bază ortogonală

$L(v_1, \dots, v_k) = L(u_1, \dots, u_k)$

II. Etapa de normalizare



Din B_1 obțin $B_2 = \{w_1, \dots, w_n\}$ a.î. $\|w_i\| = 1$, $i = \overline{1,n}$

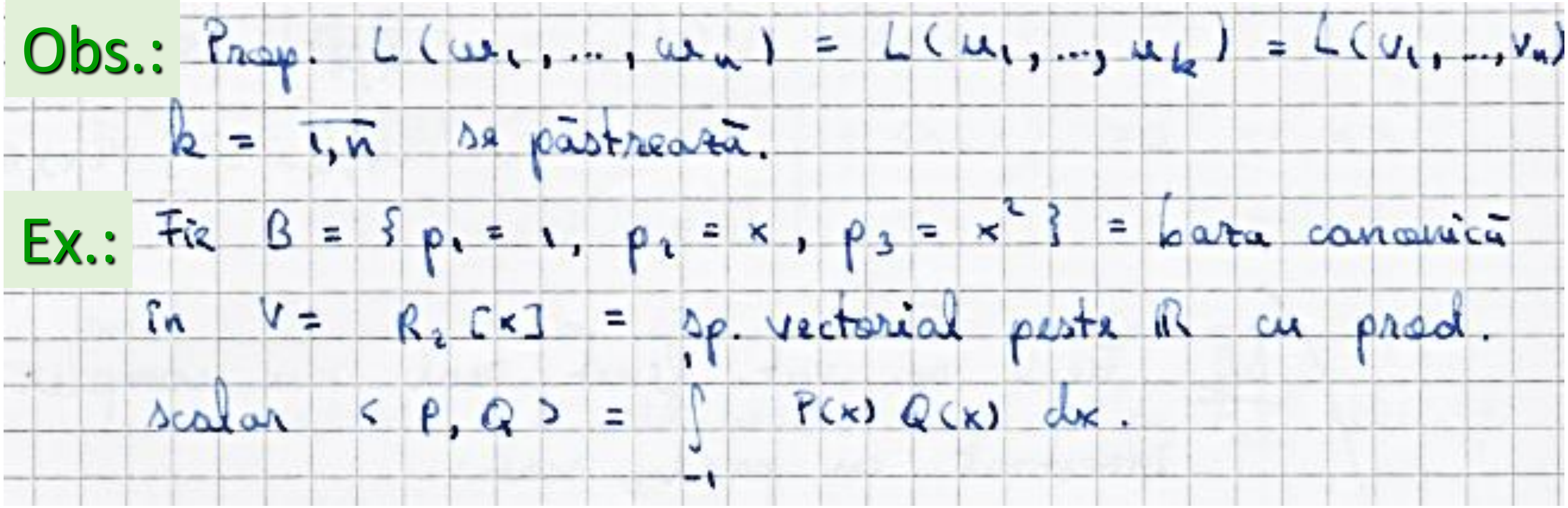
$\langle w_i, w_j \rangle = 0$

pt. $i \neq j$, $i,j = \overline{1,n}$

$$w_1 = \frac{1}{\|u_1\|} u_1$$

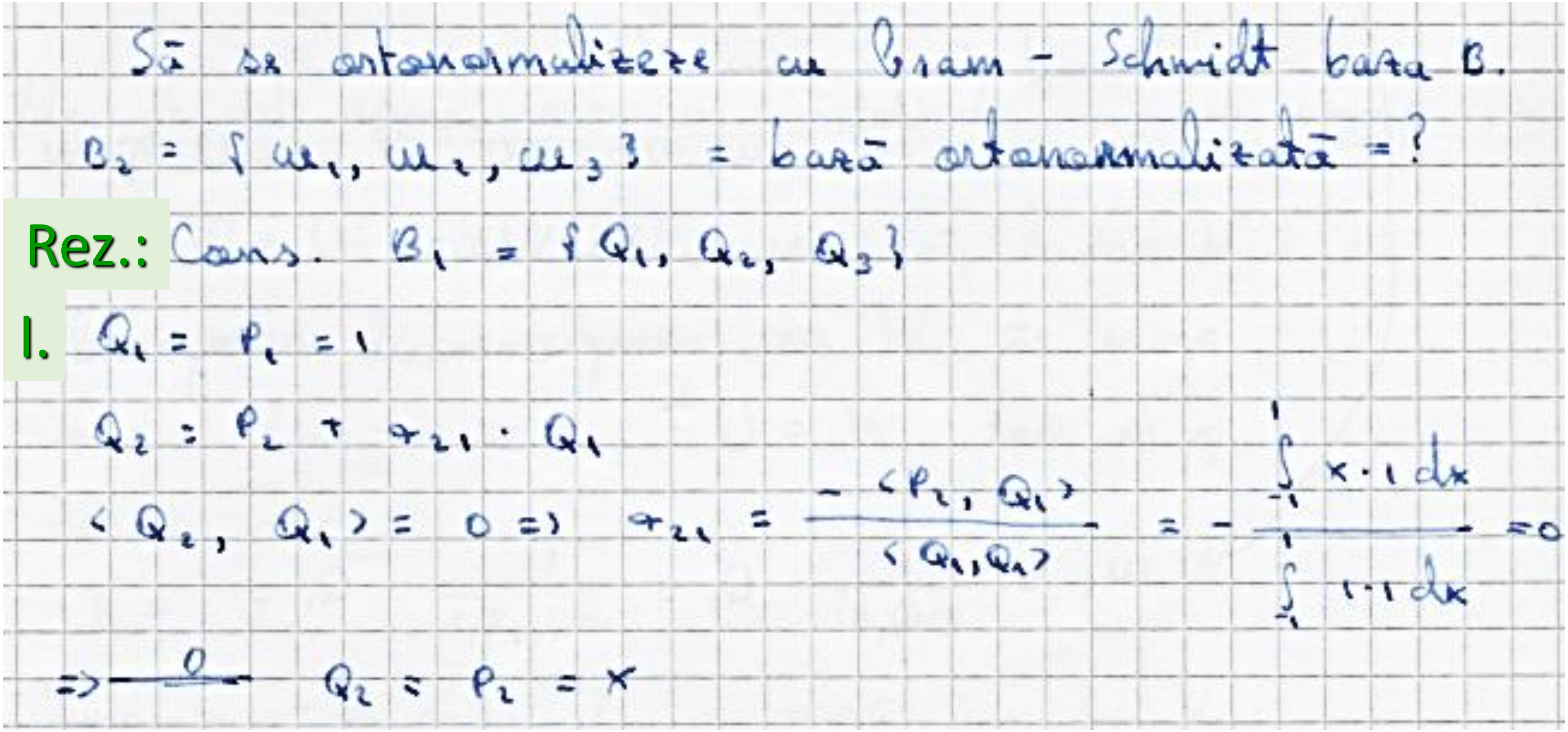
$$\vdots$$

$$w_n = \frac{1}{\|u_n\|} u_n$$



Obs.: Prop. $L(w_1, \dots, w_n) = L(u_1, \dots, u_k) = L(v_1, \dots, v_n)$
$k = \overline{1,n}$ se păstrează.

Ex.: Fie $B = \{p_1 = 1,\ p_2 = x,\ p_3 = x^2\}$ = baza canonică
în $V = R_2[x]$ = sp. vectorial peste $\mathbb{R}$ cu prod.
scalar $\langle P, Q \rangle = \int_{-1}^{1} P(x)\, Q(x)\, dx$.



Să se ortonormalizeze cu Gram - Schmidt baza B.

$B_2 = \{u_1, u_2, u_3\}$ = bază ortonormalizată = ?

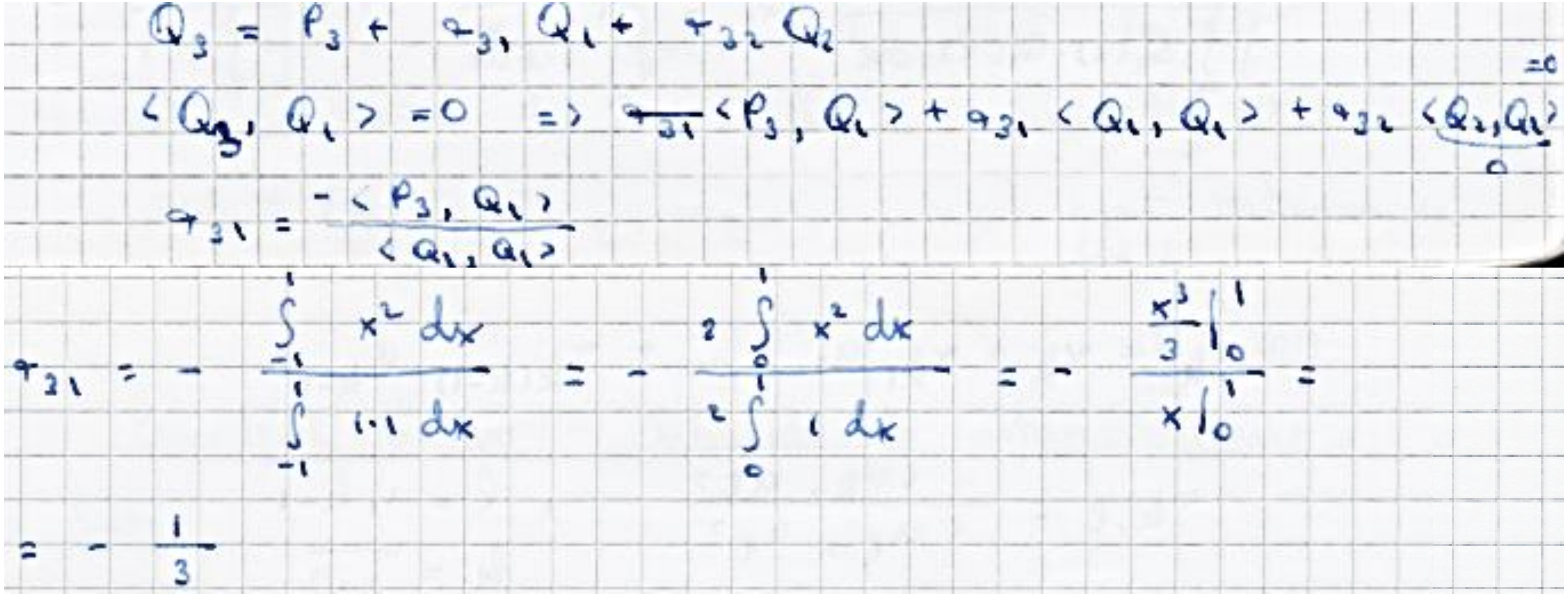
Rez.: Cons. $B_1 = \{Q_1, Q_2, Q_3\}$

I. $Q_1 = P_1 = 1$

$Q_2 = P_2 + \alpha_{21} \cdot Q_1$

$$\langle Q_2, Q_1 \rangle = 0 \Rightarrow \alpha_{21} = \frac{-\langle P_2, Q_1 \rangle}{\langle Q_1, Q_1 \rangle} = -\frac{\int_{-1}^{1} x \cdot 1\, dx}{\int_{-1}^{1} 1 \cdot 1\, dx} = 0$$

$\Rightarrow \underline{\quad 0 \quad}$ $Q_2 = P_2 = x$



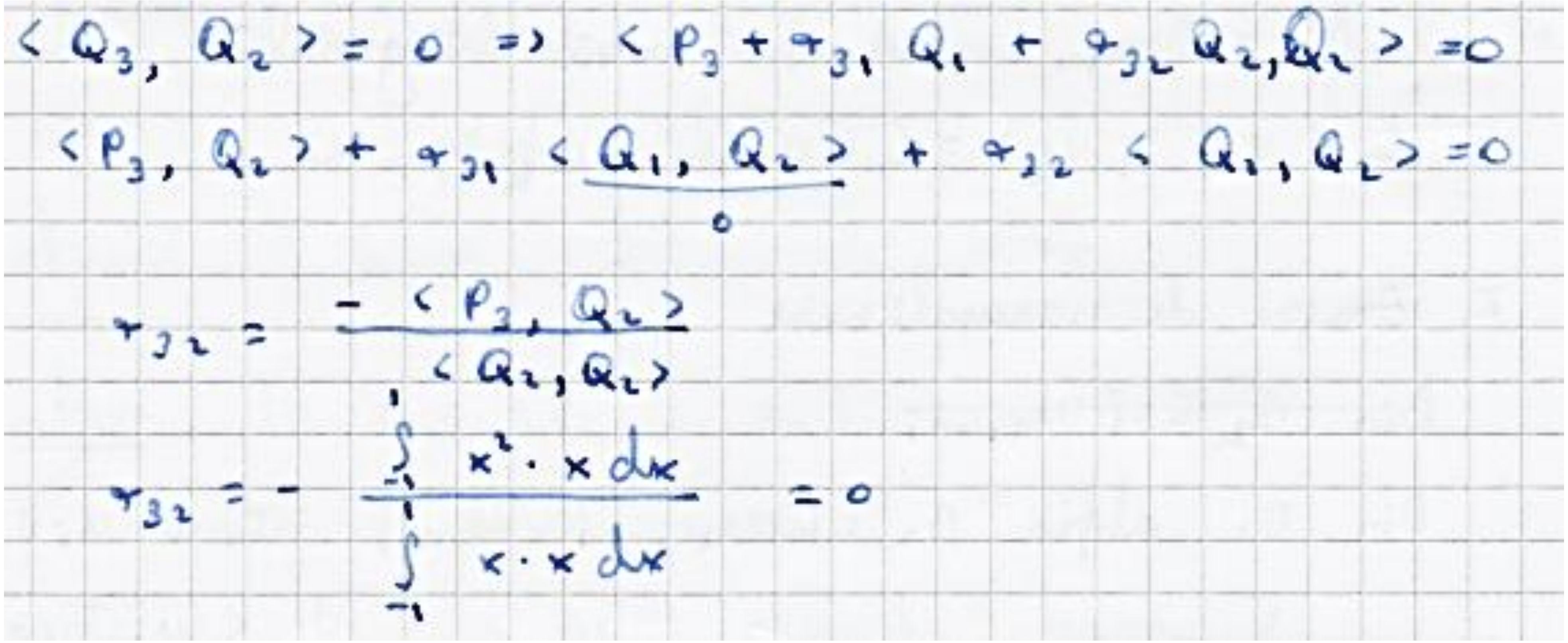
$$Q_3 = P_3 + a_{31} Q_1 + a_{32} Q_2$$

$$\langle Q_3, Q_1 \rangle = 0 \Rightarrow \cancel{a_{21}} \langle P_3, Q_1 \rangle + a_{31} \langle Q_1, Q_1 \rangle + a_{32} \underbrace{\langle Q_2, Q_1 \rangle}_{0} = 0$$

$$a_{31} = \frac{-\langle P_3, Q_1 \rangle}{\langle Q_1, Q_1 \rangle}$$

$$a_{21} = - \frac{\int_{-1}^{1} x^2 \, dx}{\int_{-1}^{1} 1 \cdot 1 \, dx} = - \frac{2 \int_{0}^{1} x^2 \, dx}{2 \int_{0}^{1} 1 \, dx} = - \frac{\frac{x^3}{3} \Big|_0^1}{x \Big|_0^1} =$$

$$= - \frac{1}{3}$$

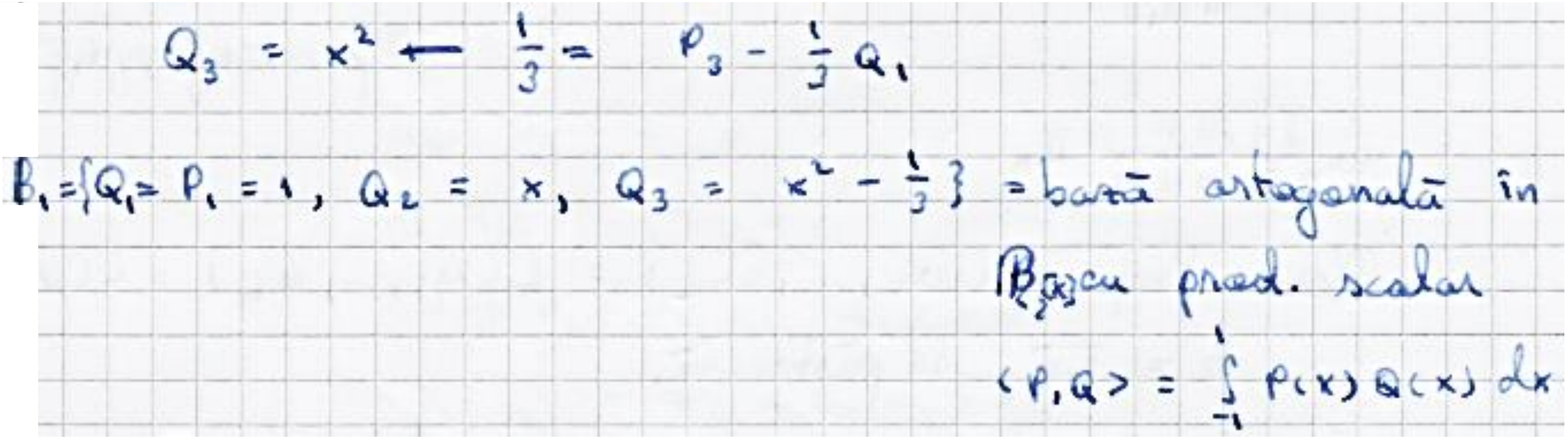


$$\langle Q_3, Q_2 \rangle = 0 \Rightarrow \langle P_3 + \alpha_{31} Q_1 + \alpha_{32} Q_2, Q_2 \rangle = 0$$

$$\langle P_3, Q_2 \rangle + \alpha_{31} \underbrace{\langle Q_1, Q_2 \rangle}_{0} + \alpha_{32} \langle Q_2, Q_2 \rangle = 0$$

$$\alpha_{32} = \frac{- \langle P_3, Q_2 \rangle}{\langle Q_2, Q_2 \rangle}$$

$$\alpha_{32} = - \frac{\int_{-1}^{1} x^2 \cdot x \, dx}{\int_{-1}^{1} x \cdot x \, dx} = 0$$



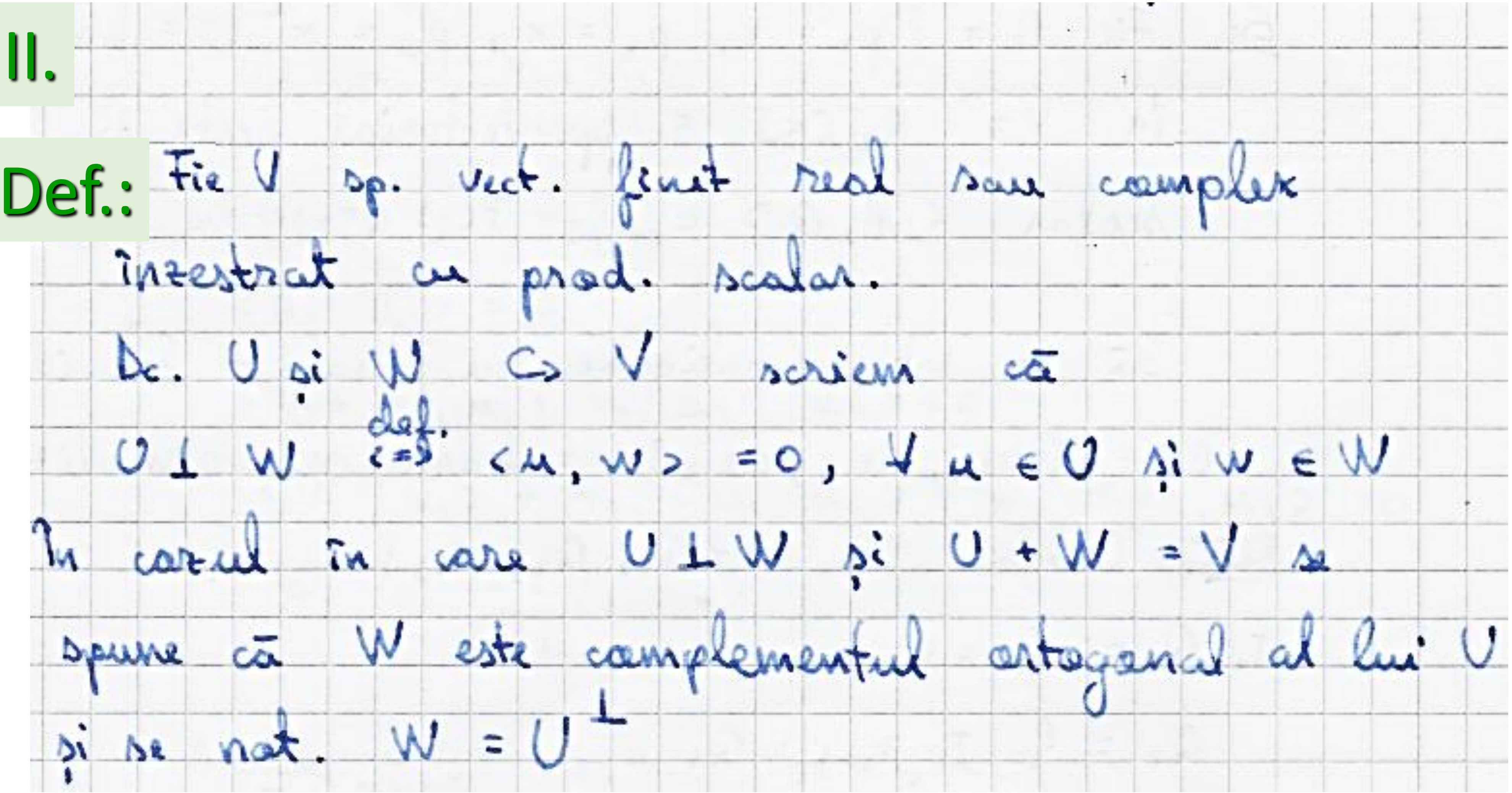
$$Q_3 = x^2 \longleftarrow \frac{1}{3} = P_3 - \frac{1}{3} Q_1$$

$B_1 = \{Q_1 = P_1 = 1,\ Q_2 = x,\ Q_3 = x^2 - \frac{1}{3}\}$ = bază ortogonală în $\mathbb{R}_2[x]$ cu prod. scalar

$$\langle P, Q \rangle = \int_{-1}^{1} P(x)\, Q(x)\, dx$$

II.

Def.:

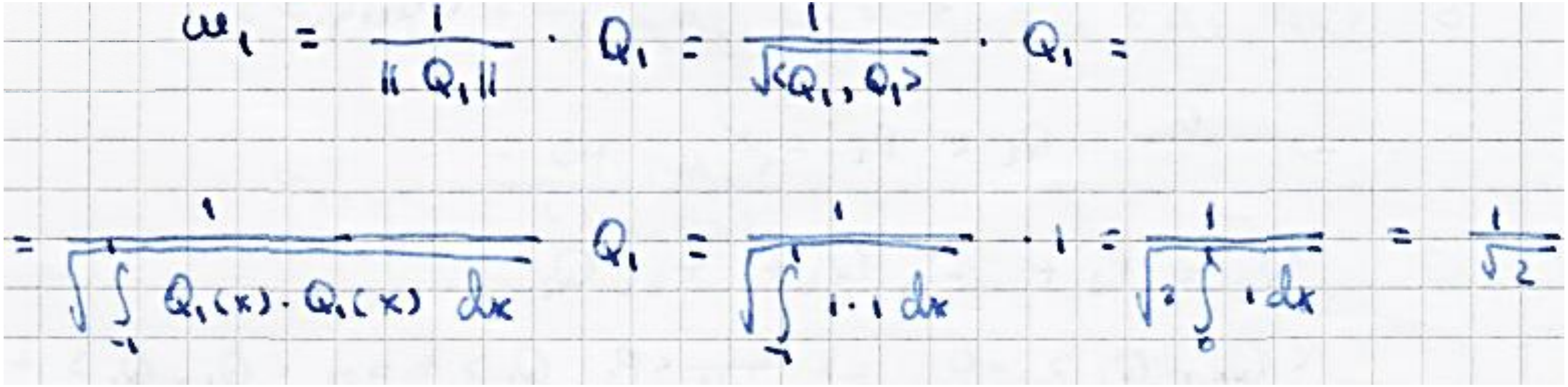


Fie V sp. vect. finit real sau complex înzestrat cu prod. scalar.

Dc. U și $W \hookrightarrow V$ scriem că

$$U \perp W \overset{def.}{\iff} \langle u, w \rangle = 0, \ \forall\, u \in U \text{ și } w \in W$$

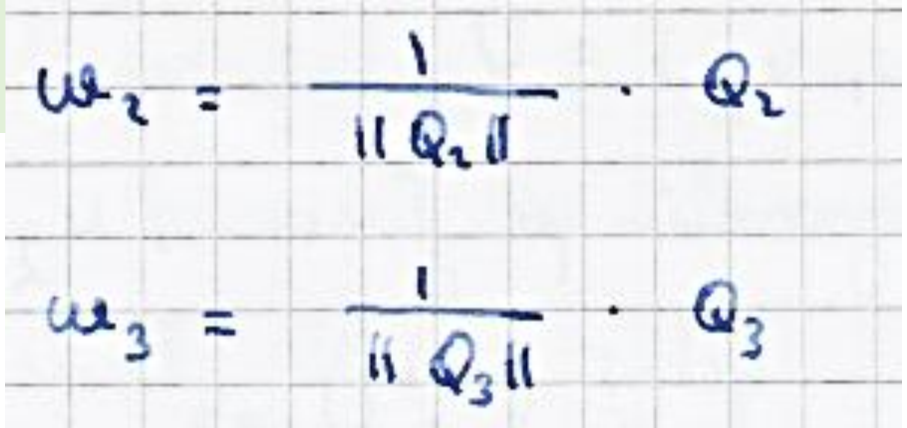
În cazul în care $U \perp W$ și $U + W = V$ se spune că W este complementul ortogonal al lui U și se not. $W = U^{\perp}$



$$w_1 = \frac{1}{\| Q_1 \|} \cdot Q_1 = \frac{1}{\sqrt{\langle Q_1, Q_1 \rangle}} \cdot Q_1 =$$

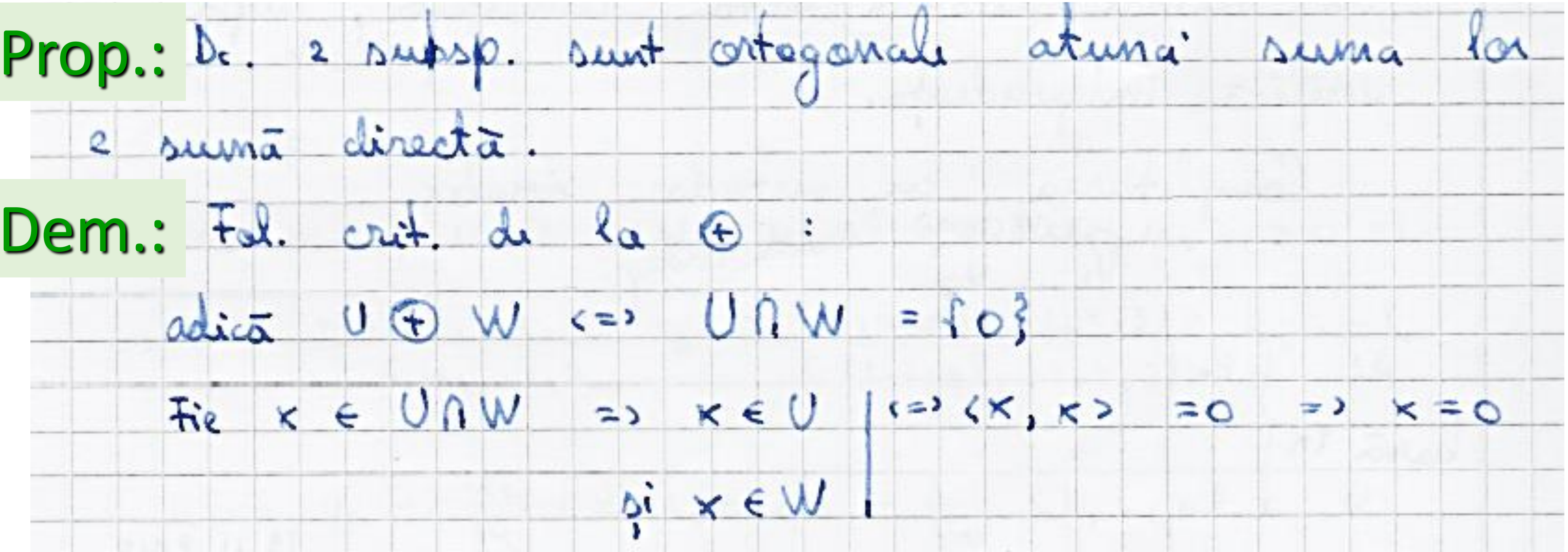
$$= \frac{1}{\sqrt{\int_{-1}^{1} Q_1(x) \cdot Q_1(x)\, dx}}\, Q_1 = \frac{1}{\sqrt{\int_{-1}^{1} 1 \cdot 1\, dx}} \cdot 1 = \frac{1}{\sqrt{2 \int_{0}^{1} 1\, dx}} = \frac{1}{\sqrt{2}}$$

Temă:



$$w_2 = \frac{1}{\| Q_2 \|} \cdot Q_2$$

$$w_3 = \frac{1}{\| Q_3 \|} \cdot Q_3$$



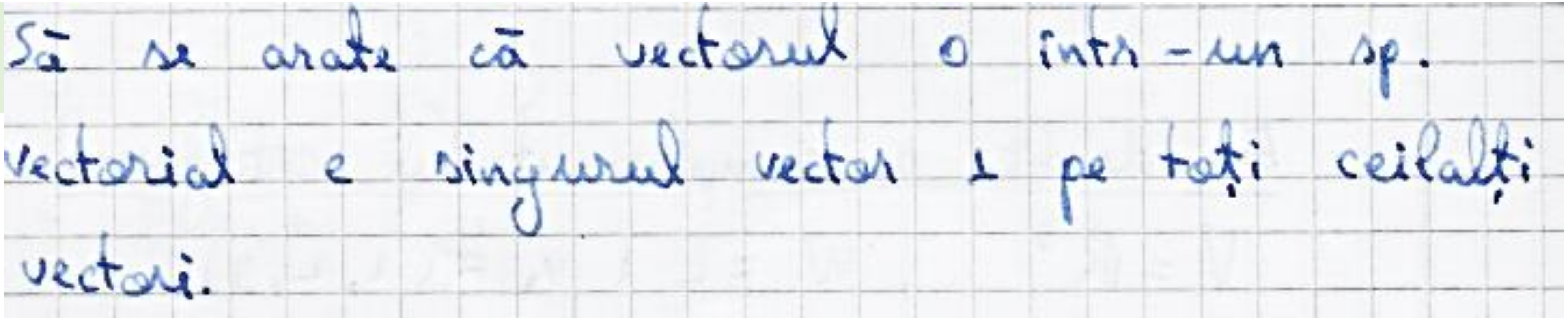
Prop.: Dc. 2 subsp. sunt ortogonale atunci suma lor e sumă directă.

Dem.: Fol. crit. de la $\oplus$:

adică $U \oplus W \Leftrightarrow U \cap W = \{0\}$

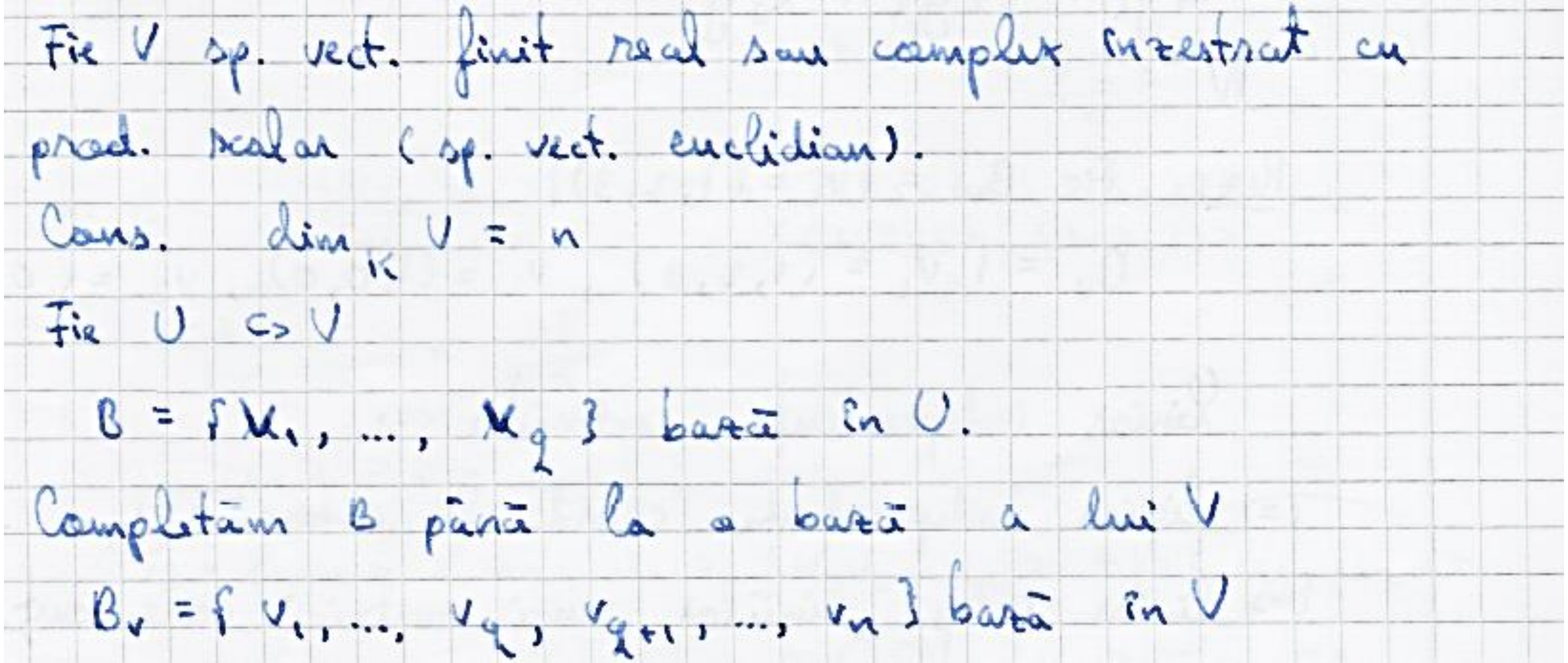
$$\text{Fie } x \in U \cap W \Rightarrow \left.\begin{array}{l} x \in U \\ \text{și } x \in W \end{array}\right| \Leftrightarrow \langle x, x \rangle = 0 \Rightarrow x = 0$$

Temă:



Să se arate că vectorul 0 intr-un sp. vectorial e singurul vector $\perp$ pe toți ceilalți vectori.

Algoritm pentru determinarea complementului ortogonal al unui subspațiu vectorial



Fie V sp. vect. finit real sau complex înzestrat cu prod. scalar (sp. vect. euclidian).

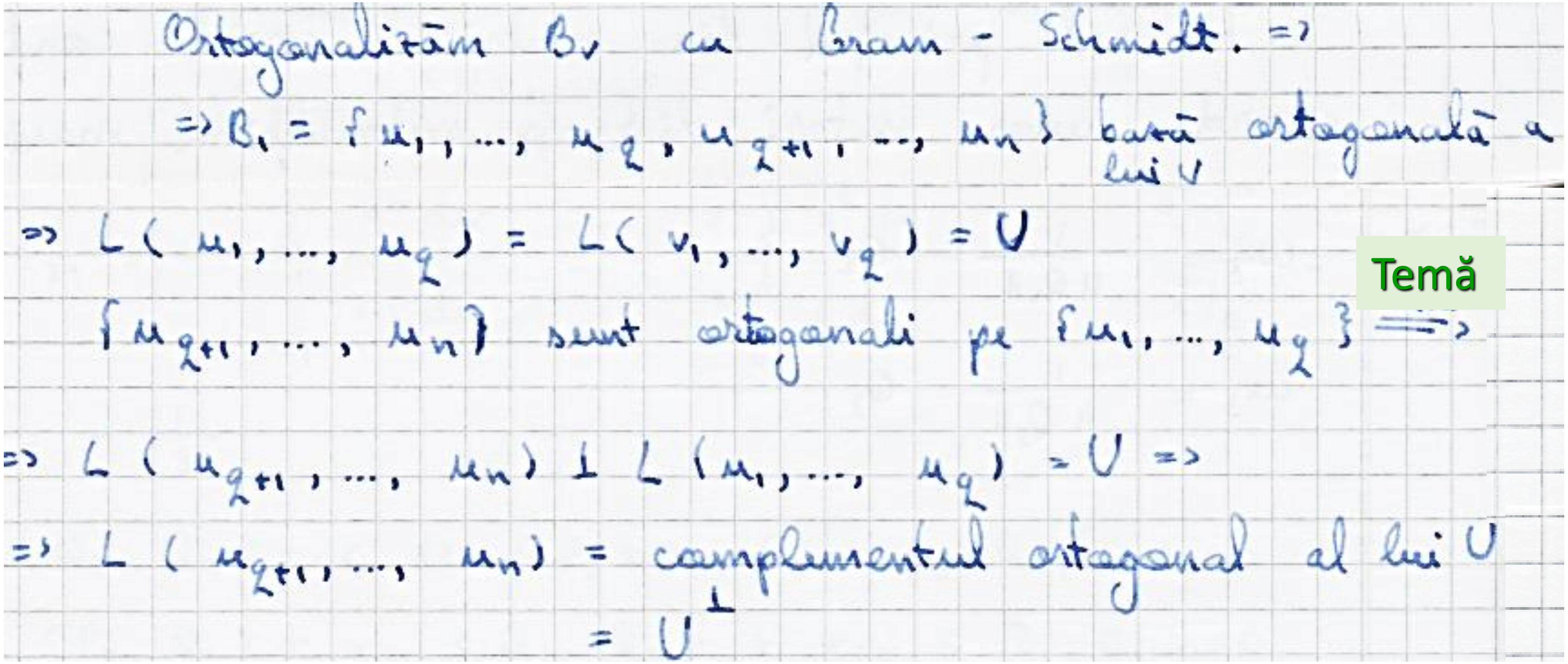
Cons. $\dim_K V = n$

Fie $U \hookrightarrow V$

$B = \{x_1, \dots, x_q\}$ bază în U.

Completăm B până la o bază a lui V

$B_V = \{v_1, \dots, v_q, v_{q+1}, \dots, v_n\}$ bază în V



Ortogonalizăm B_v cu Gram - Schmidt. $\Rightarrow$

$\Rightarrow B_1 = \{u_1, \dots, u_q, u_{q+1}, \dots, u_n\}$ bază ortogonală a lui V

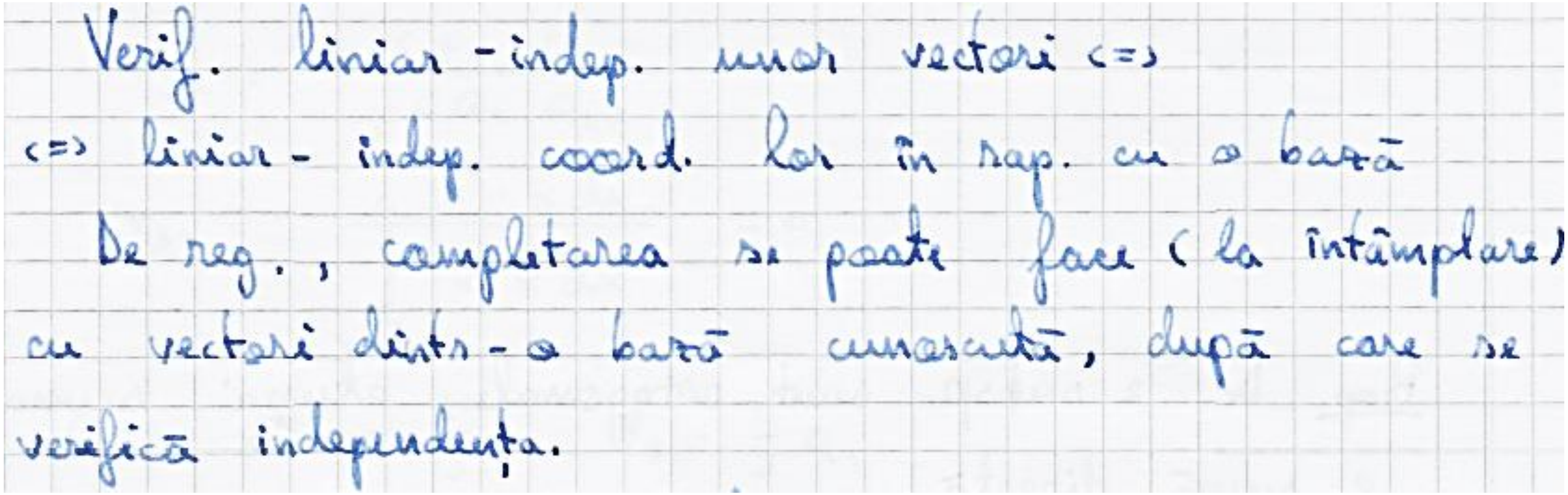
$\Rightarrow L(u_1, \dots, u_q) = L(v_1, \dots, v_q) = U$

$\{u_{q+1}, \dots, u_n\}$ sunt ortogonali pe $\{u_1, \dots, u_q\} \Longrightarrow$

$\Rightarrow L(u_{q+1}, \dots, u_n) \perp L(u_1, \dots, u_q) = U \Rightarrow$

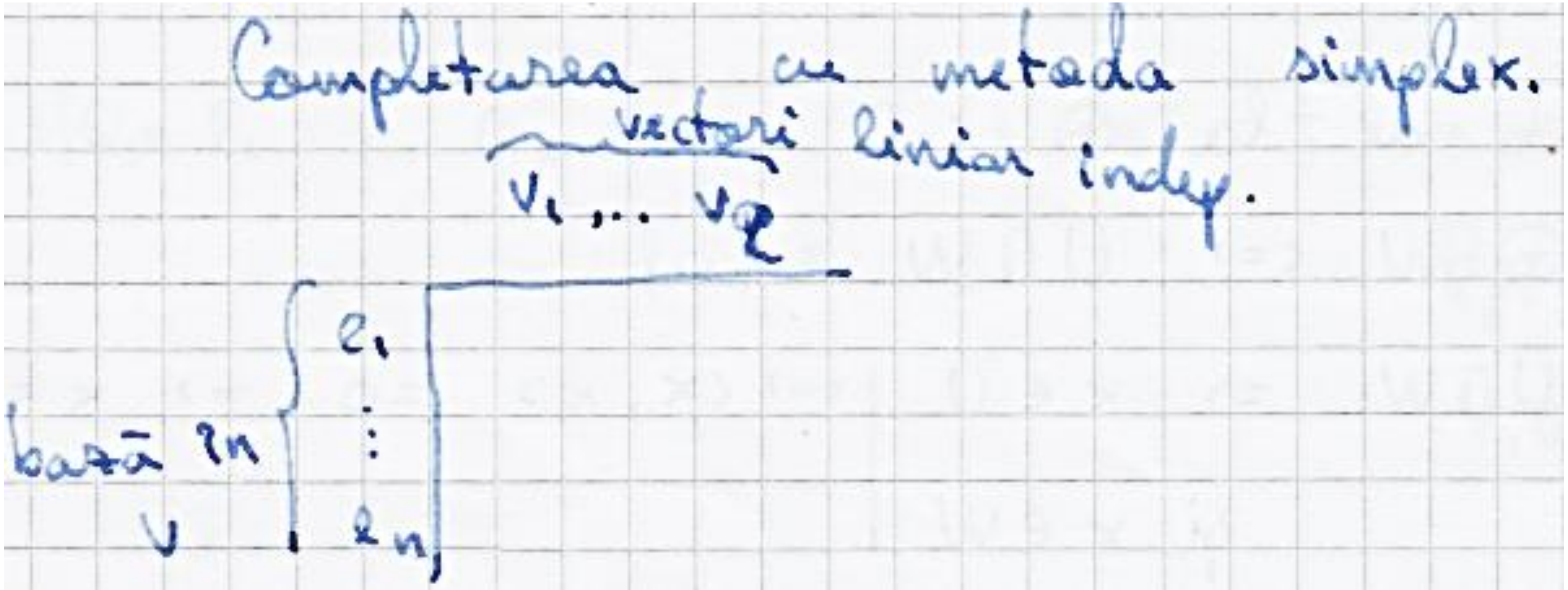
$\Rightarrow L(u_{q+1}, \dots, u_n) =$ complementul ortogonal al lui U $= U^{\perp}$

Temă



Verif. liniar-indep. unor vectori $\Leftrightarrow$
$\Leftrightarrow$ liniar-indep. coord. lor în rap. cu o bază

De reg., completarea se poate face (la întâmplare)
cu vectori dintr-o bază cunoscută, după care se
verifică independența.



Completarea cu metoda simplex.

$$\overbrace{v_1, \dots v_l}^{\text{vectori liniar indep.}}$$

$$\text{bază în } V \left\{ \begin{array}{c|l} & \\ \hline e_1 & \\ \vdots & \\ e_n & \end{array} \right.$$